

Fit a Fitted Sheet to the Mattress SOP (1x Sheet, in situ)

One episode fits one fitted sheet to one mattress, in place on its bed frame.

The base is **passive**. It has no drive of its own. It is pushed by hand to the near long side of the bed and locked there. Nothing is carried away to a table. Everything the episode touches is already at the bed when recording starts: the mattress on its frame, and the sheet lying spread on top of it, pockets down and not yet hooked.

The bed is worked **as found**. The mattress is small enough that both grippers reach every corner of it from the one park. The episode ends with all four corner pockets hooked and seated, the sheet pulled taut, the top face smooth, and the elastic hem tucked under the bottom edge all the way round.

This is an in-situ task. Three rules follow from that.

First, **the mattress and the bed frame never move**. The mattress is never lifted, never dragged, and never pushed along the frame. No gripper, wrist, or forearm rests on the frame or presses on the mattress hard enough to shift it. No pull on the sheet is ever hard enough to move the mattress.

Second, **the sheet is worked on the mattress**. It is never lifted off as a whole and never carried. A corner is lifted only as far as it takes to hook it. Everything else is done with the cloth resting on the mattress.

Third, **a far corner is reached across the top face, low and level**. The arm passes over the sheet and never rests on it. No arm swings high over the mattress and no arm reaches around an end of the bed.

The order never changes. Hook the far-left corner. Hook the far-right corner. Hook the near-left corner. Hook the near-right corner. Tension the sheet. Smooth the center. Tuck the hems. Each step is finished and checked before the next one starts.

One corner at a time, one gripper hooking, the other pinning. While one gripper hooks a corner, the other gripper pins the sheet flat on its own half of the top face, so the sheet does not drag toward the corner being hooked.

The two arms never cross. The **left gripper** works the left half of the mattress: the far-left corner, the near-left corner, the left end, and the left half of every hem. The **right gripper** works the right half: the far-right corner, the near-right corner, the right end, and the right half of every hem. The left gripper always stays left of the right gripper.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the **near long side** of the bed. Stop it **square to the bed**, so the near edge of the mattress runs straight across the frame and neither end of it sits nearer than the other.
2. Stop it **centered on the mattress**, so the left end and the right end of the mattress are the same distance out from the middle of the base.
3. Stop it **close enough** that both grippers reach the far edge of the mattress, and both far corners, low and level across the top face without an arm extending. Stop it **far enough** that no arm, wrist, or part of the base touches the bed frame or the near side face of the mattress while both arms work.
4. Check the **height band**. With the base parked, both grippers reach down past the **bottom edge** of the mattress on the near side, so a hem can be pushed under it. Both grippers also reach the top face of the mattress level, and reach over the far edge and down the far face far enough to pull a hem under it.
5. Check the **left lane**. The **left gripper** reaches the far-left corner, the near-left corner, the middle of the left end, and the left half of the far and near edges, all without extending and without reaching right of the middle of the base.
6. Check the **right lane**. The **right gripper** reaches the far-right corner, the near-right corner, the middle of the right end, and the right half of the far and near edges, all without extending and without reaching left of the middle of the base.
7. Check the **arm path** to each far corner. The gripper reaches the far corner with the arm passing over the top face, not resting on it, and without the wrist or forearm touching the mattress on the way out or back.
8. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. Do not work a mattress the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bed hard enough to shift it. It is never repositioned mid-task. A base that moves after

recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the mattress. Its frame includes the whole top face, all four corners, the near side face down to the bottom edge, and both ends.
3. The camera reads the top face from the near side. You can see whether the sheet lies square, whether a corner pocket is hooked, whether the top is taut or wrinkled, and whether any hem lies on top of the mattress or hangs loose outside it.
4. The camera reads the far corners. You can see the elastic pulled tight over each far corner and the far edge of the sheet lying flat along the far edge of the mattress.
5. Both arms are at home with grippers open.
6. The **left arm** reaches the far-left corner, the near-left corner, the left end, and the left half of the far and near edges without extending to a joint limit. It never needs to reach the right half.
7. The **right arm** reaches the far-right corner, the near-right corner, the right end, and the right half of the far and near edges without extending to a joint limit. It never needs to reach the left half.
8. Each gripper reaches a far corner low and level across the top face, and comes back the same way. Neither wrist nor forearm rests on the sheet on the way.
9. The two arms do not touch each other. The left gripper stays left of the right gripper throughout.
10. If a place cannot be reached, re-park the base by the Base positioning steps until lines 6 to 9 hold.

Materials checklist

1. The **bed frame** stands where it lives. It is not moved, not leaned on, and not pushed at any point.
2. The **mattress** is a small mattress, cot size, lying on the frame with all four corners square and firm enough that a corner holds its shape when a pocket is pulled over it. **Unvalidated.**
3. The mattress does not slide on the frame. Push once by hand on its near side face before recording: it must not shift.
4. The **top face** of the mattress and all four **side faces** are open to a gripper. Nothing rises above the top face on any side: no rail, no headboard, no footboard above the top face. There is clear space beside every side face for a gripper to reach the **bottom edge** and push a hem under it.
5. The mattress is bare. Nothing lies on it except the sheet. No mattress protector, pad, or second sheet.

6. One **fitted sheet**, cot size, cut to fit this mattress. Its **elastic hem** runs all the way round, unbroken, with no loose thread and no place where the elastic has pulled out of the seam.
7. All four **corner pockets** are open and hold their shape. Each mouth stands open when its hem is lifted, instead of staying shut.
8. The sheet is dry, clean, and pressed, with no tears and no open seams. It is not so worn that the elastic has gone slack and a hooked corner lets go on its own.
9. The sheet lies **spread on the top face, pockets down**, centered on the mattress by eye, so each corner pocket hangs at its own mattress corner and the hem lies evenly all round. **No corner is hooked.** No corner is folded under the sheet. No part of the sheet hangs so far that it touches the floor.
10. The floor around the bed and the space between the base and the near side face are clear, so an arm reaching down to the bottom edge passes over nothing.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye against the bed itself: the four edges of the mattress, the four corners, and the middle of the top face.

- **Bed frame:** the frame the mattress lies on. It is never moved, never leaned on, and never pushed.
- **Mattress:** the one mattress worked. All sheet work happens on it and around its four sides.
- **Top face:** the flat top of the mattress. The sheet lies here, and every pin and every stroke happens here.
- **Near edge:** the long edge of the mattress nearest the base. The **near side face** runs down from it to the **bottom edge**.
- **Far edge:** the long edge of the mattress furthest from the base. The **far side face** runs down from it.
- **Left end** and **right end:** the two short ends of the mattress, named as seen from the base.
- **Near-left, near-right, far-left, and far-right corner:** the four mattress corners, named from the base.
- **Left half** and **right half:** the top face split down its middle from the near edge to the far edge. Everything left of that line is the **left gripper's**. Everything right of it is the **right gripper's**.

Arm lanes

The two arms never cross. This holds for the whole episode.

- The **left gripper works the left lane:** the far-left corner, the near-left corner, the left end, and the left half of the far edge and near edge.
- The **right gripper works the right lane:** the far-right corner, the near-right corner, the right end, and the right half of the far edge and near edge.

- The **left gripper always stays left of the right gripper**. Neither arm reaches over, under, around, or past the other. Neither arm reaches across the front of the other arm's body.
- When one gripper hooks a corner, the other gripper pins on its own half of the top face, near the middle. It never pins on the corner being hooked.
- Nothing is handed between grippers. Only one corner is hooked at a time.

Arm assignments

- **Left gripper.** Hooks the far-left and near-left corners. Pins the left half while the right gripper hooks. Pulls the left end in the tension. Smooths the left half. Tucks the left half of the far and near hems and the left end.
- **Right gripper.** Hooks the far-right and near-right corners. Pins the right half while the left gripper hooks. Pulls the right end in the tension. Smooths the right half. Tucks the right half of the far and near hems and the right end.

Vocabulary

- **Pockets down:** the sheet lies with its four corner pockets facing the mattress, so each pocket hangs at a mattress corner ready to go over it. The sheet is staged this way and stays this way.
- **Corner pocket:** the sewn corner of a fitted sheet. It is a cup of cloth with an opening, and the opening goes over a mattress corner.
- **Pocket mouth:** the opening of a corner pocket. It is what goes over the corner.
- **Seam point:** the sewn tip of a corner pocket, where the two side seams meet. It is the only part of a pocket a gripper pinches when hooking it.
- **Elastic hem:** the gathered band running all the way round the edge of the sheet. It bunches into a wavy edge when it is loose, and pulls straight when it is stretched.
- **Hook:** the gripper pinches a corner pocket at its seam point, lifts it clear, carries it out past the mattress corner, lowers it so the pocket mouth goes over the top of the corner, and keeps pulling straight down the corner until the hem passes under the bottom edge. A hook is one continuous pull. It is not a push, and the pocket is never shoved onto the corner with an open gripper. **Unvalidated.**
- **Hooked:** the pocket covers the top of the corner and both side faces at that corner, and its hem sits under the bottom edge. Nothing of the pocket lies on the top face.
- **Seated:** a hooked pocket with its elastic pulled tight over the corner, so the cloth lies smooth down the corner with no fold and no gap between the cloth and the mattress. A hooked pocket that is loose or bunched is not seated.
- **Popped:** a pocket that was hooked and has let go of the corner, so any part of its mouth is back on top of the corner or off the corner.

- **Pin:** the gripper closes, comes straight down, and presses on the sheet to hold it still on the top face. It stays there without moving until the step says it can lift. A pin does not squeeze the cloth between the fingers and it does not slide along.
- **Lift clear:** raise the cloth straight up until there is daylight under it, before anything moves sideways.
- **Stroke:** the gripper closes, presses down on the sheet, and runs along the top face in one smooth line from where it starts to an edge, staying in contact the whole way. It never lifts off in the middle and it never runs past the edge.
- **Taut:** the top of the sheet lies flat and tight on the top face, with no slack to lift between two fingers, and no wrinkle running from one corner toward another.
- **Bite:** one pinch of the hem, pull down the side face, and push under the bottom edge, worked at one spot along an edge, then the gripper opens.
- **Tucked:** the hem at that spot lies under the bottom edge, out of sight from the side, with no cloth hanging outside the side face and none lying on the top face.
- **Resting:** the cloth is down and takes its own weight. It stays put when the gripper lifts clear.
- **Release point:** with the cloth resting on the mattress, or with a hem already under the bottom edge. A gripper opens nowhere else. Nothing is let go in the air.

Steps

Run Steps 1 to 7 once, in order. One sheet is on the mattress for the whole episode and it never comes off.

Step 1: Check and square the spread sheet

Goal: the sheet lies spread on the top face, pockets down, centered by eye, with each corner pocket hanging at its own mattress corner and nothing hooked.

- Look at the sheet on the mattress. It lies pockets down, centered, with all four corner pockets hanging at their corners and the hem lying evenly all round.
- If a corner is folded under the sheet, the gripper on that side pinches it at its **seam point**, lifts it clear, carries it out, and lets it hang at its corner.
- If the sheet sits crooked or off center, the **right gripper** pinches the near-right corner at its seam point and the **left gripper** pinches the near-left corner at its seam point. Both lift clear together and lay the sheet down again so it sits centered. Both open at the same time, with the cloth resting.
- Both grippers lift away.

Square by lifting and laying down. Never drag the sheet across the top face to square it.

Check: the sheet lies spread and centered, pockets down. Each corner pocket hangs at its own corner. No corner is folded under. Nothing is hooked. If the sheet is bunched, lying pockets up,

or already hooked at any corner, stop the episode and stage it again from the reset list.

Step 2: Hook the far corners

Goal: both far corner pockets are hooked and seated, with the sheet still lying spread on the top face.

The far corners are hooked first, while the sheet is slack. One gripper hooks. The other gripper pins.

2.1 Hook the far-left corner

- With the **right gripper** closed, come straight down and **pin** the sheet on the **right half** of the top face, near the middle. Hold the pin.
- With the **left gripper** open, reach across the **left half** of the top face, low and level, to the far-left corner. Keep the arm above the sheet. Do not rest the wrist or forearm on it.
- Pinch the far-left corner pocket at its **seam point**. Lift it clear.
- Carry it out past the far-left corner of the mattress, so the pocket mouth hangs open over the top of the corner.
- **Hook** it: lower the pocket over the corner and keep pulling straight down the corner until the hem passes under the bottom edge. One continuous pull.
- Open the **left gripper** only when the hem is under the bottom edge. Draw the **left gripper** back across the left half, low and level.
- Lift the **right gripper** pin straight up.

Check: the far-left pocket is hooked and seated. The elastic reads tight over the corner and nothing of the pocket lies on the top face. If it is loose or bunched, the **left gripper** pinches its hem just beside the corner and pulls it further down and under. If it has popped, hook it again with the **left gripper**, with the **right gripper** pinning.

2.2 Hook the far-right corner

- With the **left gripper** closed, come straight down and **pin** the sheet on the **left half** of the top face, near the middle. Hold the pin.
- With the **right gripper** open, reach across the **right half** of the top face, low and level, to the far-right corner. Keep the arm above the sheet.
- Pinch the far-right corner pocket at its **seam point**. Lift it clear. Carry it out past the far-right corner so the pocket mouth hangs open over the top of the corner.
- **Hook** it in one continuous pull down the corner until the hem passes under the bottom edge.
- Open the **right gripper** only when the hem is under the bottom edge. Draw the **right gripper** back across the right half, low and level.
- Lift the **left gripper** pin straight up.

Check: both far pockets are hooked and seated, and the far edge of the sheet lies along the far edge of the mattress. If the far-right pocket is loose, the **right gripper** pinches its hem beside the corner and pulls it down and under. If either far pocket has popped, hook it again with the gripper for that side, with the other gripper pinning.

Expected state: two far corners hooked, two near corners still hanging loose, the sheet lying spread on the top face with slack toward the near edge.

Step 3: Hook the near corners

Goal: all four corner pockets are hooked and seated.

The near corners take the stretch. Each near pocket is pulled toward the base and down over its corner against the pull of the far pockets already hooked.

3.1 Hook the near-left corner

- With the **right gripper** closed, come straight down and **pin** the sheet on the **right half** of the top face, near the middle. Hold the pin.
- With the **left gripper** open, pinch the near-left corner pocket at its **seam point**. Lift it clear.
- Carry it out past the near-left corner of the mattress, pulling gently toward the base, so the pocket mouth hangs open over the top of the corner.
- **Hook** it in one continuous pull down the corner until the hem passes under the bottom edge.
- Open the **left gripper** only when the hem is under the bottom edge. Draw the **left gripper** away.
- Lift the **right gripper** pin straight up.

Check: the near-left pocket is hooked and seated. The far-left pocket is still hooked. If the near-left pocket is loose, the **left gripper** pinches its hem beside the corner and pulls it down and under. If the far-left pocket has popped, hook it again with the **left gripper**, with the **right gripper** pinning, before going on.

3.2 Hook the near-right corner

- With the **left gripper** closed, come straight down and **pin** the sheet on the **left half** of the top face, near the middle. Hold the pin.
- With the **right gripper** open, pinch the near-right corner pocket at its **seam point**. Lift it clear.
- Carry it out past the near-right corner of the mattress, pulling gently toward the base, so the pocket mouth hangs open over the top of the corner.
- **Hook** it in one continuous pull down the corner until the hem passes under the bottom edge.
- Open the **right gripper** only when the hem is under the bottom edge. Draw the **right gripper** away.
- Lift the **left gripper** pin straight up.

Check: all four pockets are hooked and seated. The elastic reads tight over every corner. If any pocket is loose, the gripper for that side pinches its hem beside the corner and pulls it down and under. If any pocket has popped, hook it again with the gripper for that side, with the other gripper pinning.

Expected state: four corners hooked, the sheet covering the whole top face, some slack and wrinkles still showing in the middle, and loose hem showing along the edges.

Step 4: Tension the sheet

Goal: the sheet is pulled taut along the length of the mattress, and all four pockets stay hooked.

Both grippers work at the same time. They pull in opposite directions, away from each other, so the sheet is stretched between them and neither end drags the other.

- With the **left gripper** open, reach to the **middle of the left end** and pinch the elastic hem where it hangs at the left end face.
- With the **right gripper** open, reach to the **middle of the right end** and pinch the elastic hem where it hangs at the right end face.
- Together, pull both hems **outward**, away from each other, then **down** the end face, then **under** the bottom edge. One smooth pull each, at the same time. Pull only until the top reads taut. Do not pull hard enough to shift the mattress or pop a corner.
- Open both grippers only when each hem is under its bottom edge. Lift both grippers away.

Check: the top of the sheet is taut. All four pockets are still hooked and seated. The mattress has not shifted. If a pocket has popped, hook it again with the gripper for that side, with the other gripper pinning, then run the tension pull once more. If the top still lifts slack between two fingers, run the tension pull once more.

Expected state: four corners hooked and seated, the sheet taut end to end, wrinkles in the middle only, loose hem still showing along the far and near edges.

Step 5: Smooth the center

Goal: the top face is smooth, with the slack stroked out from the middle to the edges.

Every stroke starts near the middle of the top face and runs outward to an edge. Nothing is stroked toward the middle. The **left gripper** smooths the left half first. The **right gripper** then smooths the right half. The other arm stays clear and off the sheet while one gripper strokes.

5.1 Smooth the left half

- With the **left gripper** closed, come straight down just **left of the middle** of the top face.
- **Stroke** from there to the **far edge**. Lift.
- Come down again just left of the middle. **Stroke** from there to the **near edge**. Lift.
- Come down again just left of the middle. **Stroke** from there to the **left end**. Lift.

- Draw the **left gripper** away.

Check: the left half is smooth, with no wrinkle running from a corner toward the middle. If a wrinkle remains, the **left gripper** strokes once more from just left of the middle toward the edge that wrinkle runs to.

5.2 Smooth the right half

- With the **right gripper** closed, come straight down just **right of the middle** of the top face.
- **Stroke** from there to the **far edge**. Lift.
- Come down again just right of the middle. **Stroke** from there to the **near edge**. Lift.
- Come down again just right of the middle. **Stroke** from there to the **right end**. Lift.
- Draw the **right gripper** away.

Check: the whole top face is smooth and taut. All four pockets are still hooked and seated. If a wrinkle remains on the right half, the **right gripper** strokes once more from just right of the middle toward the edge that wrinkle runs to. If a pocket has popped, hook it again with the gripper for that side, with the other gripper pinning, then stroke that half again.

Expected state: four corners hooked and seated, the top face smooth and taut, loose hem showing along the far and near edges and maybe at an end.

Step 6: Tuck the loose hems

Goal: the elastic hem is tucked under the bottom edge all the way round, with no cloth hanging outside a side face and none lying on the top face.

The far edge is tucked first, then the near edge, then any end that still shows loose hem. Each edge takes one bite from each gripper, on its own half. A bite is worked at the middle of that half.

6.1 Tuck the far hem

- With the **left gripper** open, reach across the left half, low and level, to the **middle of the left half of the far edge**. Pinch the hem where it hangs at the far side face.
- **Bite:** pull the hem down the far side face and push it under the bottom edge. Open. Draw the **left gripper** back across the left half, low and level.
- With the **right gripper** open, reach across the right half, low and level, to the **middle of the right half of the far edge**. Pinch the hem where it hangs at the far side face.
- **Bite:** pull the hem down the far side face and push it under the bottom edge. Open. Draw the **right gripper** back across the right half, low and level.

Check: the far hem is tucked along its whole length. Nothing hangs outside the far side face. If hem still shows on the left half, the **left gripper** bites it once more there. If hem still shows on the right half, the **right gripper** bites it once more there.

6.2 Tuck the near hem

- With the **left gripper** open, reach to the **middle of the left half of the near edge**. Pinch the hem where it hangs at the near side face.
- **Bite**: pull the hem down the near side face and push it under the bottom edge. Open. Draw the **left gripper** away.
- With the **right gripper** open, reach to the **middle of the right half of the near edge**. Pinch the hem where it hangs at the near side face.
- **Bite**: pull the hem down the near side face and push it under the bottom edge. Open. Draw the **right gripper** away.

Check: the near hem is tucked along its whole length. Nothing hangs outside the near side face. If hem still shows on a half, the gripper for that half bites it once more there.

6.3 Tuck a loose end

- Look at the left end. If any hem hangs outside the left end face or lies on the top face at the left end, the **left gripper** pinches it at the middle of the left end and **bites** it once. If the left end is already tucked, the **left gripper** does not touch it.
- Look at the right end. If any hem hangs outside the right end face or lies on the top face at the right end, the **right gripper** pinches it at the middle of the right end and **bites** it once. If the right end is already tucked, the **right gripper** does not touch it.

Check: the hem is tucked all the way round. All four pockets are still hooked and seated. The top face is still smooth and taut. If a bite has pulled a wrinkle into the top face, the gripper for that half strokes once from just its side of the middle toward the edge that was bitten. If a pocket has popped, hook it again with the gripper for that side, with the other gripper pinning, and tuck the hem beside it once more.

Expected state: four corners hooked and seated, hem tucked all round, top face smooth and taut. Base still locked in its park.

Step 7: End the episode

- Confirm the corners. All four pockets are hooked and seated, with the elastic tight over every corner.
- Confirm the top face. It is smooth and taut, with no wrinkle running from a corner toward the middle.
- Confirm the hem. It is tucked under the bottom edge all the way round, with no cloth hanging outside a side face and none lying on the top face.
- Confirm the mattress has not shifted, the frame has not moved, and the base has not moved.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Unhook all four corner pockets by hand and pull the hem out from under the bottom edge all round.
2. Lift the sheet off the mattress, shake it out, and lay it back on the top face pockets down, centered by eye, with each corner pocket hanging at its own mattress corner and the hem lying evenly all round. Hook nothing.
3. Check no corner is folded under and no part of the sheet touches the floor.
4. Replace the sheet when its elastic has gone slack so a hooked corner lets go on its own, when a seam has opened, when a pocket mouth no longer stands open, or when it is torn or damp.
5. Check the mattress is still square on the frame and does not shift under one push by hand on its near side face. Push it square by hand if it has moved.
6. Check nothing has fallen on the floor around the bed and the space between the base and the near side face is clear.
7. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the bed shifts in frame, the mattress changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 7. The base is parked and locked before recording and stays still for the whole episode.

- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the **left gripper** touches the far-right corner, the near-right corner, the right end, or the right half of a hem. The **right gripper** touches the far-left corner, the near-left corner, the left end, or the left half of a hem. The left gripper goes to the right of the right gripper. One arm reaches over, under, around, or past the other, or across the front of the other arm's body.
- **SOP rule broken:** Steps 1 to 6. The left gripper works the left lane, the right gripper works the right lane, and the left gripper always stays left of the right gripper.
- **Coaching note:** the middle of the mattress is the line. If you have to reach past the other arm, the wrong arm is doing the job.

Violation: Mattress or frame moved

- **Visible cue:** the mattress slides, lifts, turns, or rocks on the frame at any point. A pull on the sheet drags the mattress toward the base or toward an end. A gripper, wrist, or forearm rests on the frame, or a press on the mattress shifts it. The frame rocks or shifts.
- **SOP rule broken:** Steps 1 to 6. The mattress and the frame never move. No pull on the sheet is hard enough to move the mattress, and nothing rests on the frame.
- **Coaching note:** the sheet gives before the mattress does. Pull the cloth, not the bed. The arm holds itself up.

Violation: Arm rested on or dragged across the sheet

- **Visible cue:** a wrist or forearm rests on the sheet while a gripper reaches for a far corner or the far hem, or drags across the top face on the way out or back, so the sheet moves under it. Or an arm swings high over the mattress instead of passing low and level.
- **SOP rule broken:** Steps 2.1, 2.2, and 6.1. A far corner or the far hem is reached across the top face, low and level, with the arm above the sheet and never resting on it.
- **Coaching note:** low, level, and clear of the cloth. The sheet under your forearm is the sheet you just smoothed.

Violation: Sheet dragged to square it

- **Visible cue:** in Step 1 a gripper pulls the sheet sideways across the top face to center it, instead of lifting the near corners clear and laying the sheet down again. Or one gripper lifts a near corner and lays it alone, leaving the sheet crooked.
- **SOP rule broken:** Step 1. Square by lifting both near corners clear together and laying the sheet down again, never by dragging.
- **Coaching note:** lift, carry, lay. Dragging cloth over a mattress moves the mattress and bunches the hem.

Violation: Hooking started with the sheet not squared

- **Visible cue:** Step 2 begins with the sheet off center, a corner folded under, a corner pocket not at its own mattress corner, or a corner already hooked. Or the Step 1 check is skipped.
- **SOP rule broken:** Step 1. The sheet lies spread and centered, pockets down, no corner folded under, and nothing hooked before any corner is hooked.
- **Coaching note:** look at all four corners before the first hook. A crooked start is four crooked corners.

Violation: Wrong corner order

- **Visible cue:** a near corner is hooked before both far corners are hooked. The far-right corner is hooked before the far-left. The near-right corner is hooked before the near-left. Two corners are hooked at the same time. Or a corner is skipped and come back to.
- **SOP rule broken:** Steps 2 and 3. Hook far-left, then far-right, then near-left, then near-right, one corner at a time.
- **Coaching note:** far first, left before right, one at a time. The near corners take the stretch, so they go last.

Violation: Pocket picked wrong

- **Visible cue:** a gripper pinches a corner pocket anywhere other than its seam point: by the pocket mouth, by the hem, by a handful of the body of the sheet, or by two layers at once. Or a gripper pinches the wrong corner.
- **SOP rule broken:** Steps 1, 2, and 3. A corner pocket is pinched at its seam point and nowhere else.
- **Coaching note:** find the tip where the two seams meet and take only that. A pocket picked by its mouth goes on twisted.

Violation: Pocket hooked wrong

- **Visible cue:** a gripper pushes a pocket onto the corner with an open gripper. The hook is made in two or more pulls with a release between. The pocket mouth goes over the corner sideways or twisted. The gripper opens before the hem is under the bottom edge. Or the pocket is let go in the air.
- **SOP rule broken:** Steps 2.1, 2.2, 3.1, and 3.2. A hook is one continuous pull, mouth over the top of the corner, straight down until the hem passes under the bottom edge, and the gripper opens only then.
- **Coaching note:** over the top, straight down, past the bottom edge, then let go. One pull.

Violation: Pocket left unhooked or not seated

- **Visible cue:** a step moves on with a pocket whose mouth is still on top of the corner, whose hem is not under the bottom edge, or which is hooked but loose, bunched, or folded on the

corner.

- **SOP rule broken:** Steps 2.1, 2.2, 3.1, and 3.2. Each pocket is hooked and seated before the next corner is started, with the loose pocket pulled further down and under by the gripper for that side.
- **Coaching note:** tight over the corner, smooth down both faces, hem under. If it is loose, pull it down before you move on.

Violation: Popped corner not re-hooked

- **Visible cue:** a hooked pocket lets go of its corner during a later hook, the tension pull, a stroke, or a bite, and the episode goes on without that pocket being hooked again. Or the pocket is hooked again by the wrong gripper, or without the other gripper pinning.
- **SOP rule broken:** Steps 3, 4, 5, and 6. A popped pocket is hooked again by the gripper for that side, with the other gripper pinning, before the step goes on.
- **Coaching note:** every check counts four corners. A corner that let go is hooked again the same way it was hooked the first time.

Violation: Pin missing, misplaced, or moved

- **Visible cue:** a gripper hooks a corner while the other gripper is not pinning. The pin is placed on the wrong half, on the corner being hooked, or at an edge instead of near the middle. The pinning gripper slides, squeezes the cloth, lifts before the hook is done, or presses hard enough to dent the mattress.
- **SOP rule broken:** Steps 2.1, 2.2, 3.1, and 3.2. The other gripper pins on its own half of the top face, near the middle, and holds still until the hook is done.
- **Coaching note:** press, hold, wait. The pin is what stops the sheet chasing the corner.

Violation: Tension pulled wrong

- **Visible cue:** only one end is pulled. The two ends are pulled one after the other instead of together. A gripper pinches the hem away from the middle of its end, or pinches the body of the sheet instead of the hem. A hem is pulled up or across instead of outward, down, and under. A gripper opens with its hem not under the bottom edge. Or the pull is hard enough to pop a corner or shift the mattress.
- **SOP rule broken:** Step 4. Both grippers pinch the hem at the middle of their own end and pull outward, down, and under at the same time, only until the top reads taut.
- **Coaching note:** both ends, together, away from each other. Stop when the top is tight, not when the arm runs out.

Violation: Stroke pattern wrong

- **Visible cue:** a stroke starts anywhere other than just beside the middle of the top face. A stroke runs toward the middle instead of away from it. A stroke lifts off part way, runs past an

edge, or is made with an open gripper. The three strokes of a half are made in the wrong order, or one is missing. Or the right half is stroked before the left half is done.

- **SOP rule broken:** Steps 5.1 and 5.2. Each gripper strokes from just its side of the middle to the far edge, then to the near edge, then to its own end, closed and in contact the whole way, left half first.
- **Coaching note:** middle to far, middle to near, middle to end. Push the slack to the edge, never into the middle.

Violation: Smoothing skipped or half left wrinkled

- **Visible cue:** Step 6 begins with a half of the top face not stroked at all, or with a wrinkle running from a corner toward the middle that the check named and that was not stroked out.
- **SOP rule broken:** Steps 5.1 and 5.2. Both halves are stroked, and a wrinkle the check finds is stroked out toward the edge it runs to before the tuck begins.
- **Coaching note:** three strokes each side, then look again. A wrinkle you tuck over stays for the whole episode.

Violation: Bite worked wrong

- **Visible cue:** a gripper pushes the hem under with an open gripper, pinches the body of the sheet instead of the hem, pulls the hem up onto the top face, or opens with the hem not under the bottom edge. A bite is worked away from the middle of its half. Or a gripper bites on the wrong half.
- **SOP rule broken:** Steps 6.1, 6.2, and 6.3. A bite pinches the hem at the middle of that gripper's half, pulls it down the side face, pushes it under the bottom edge, and opens only then.
- **Coaching note:** pinch the hem, down the face, under the edge, let go. One bite, at the middle of your half.

Violation: Wrong tuck order

- **Visible cue:** the near hem is bitten before the far hem. An end is bitten before both the far and near hems are tucked. The right half of an edge is bitten before the left half. Or an end that is already tucked is bitten anyway.
- **SOP rule broken:** Steps 6.1, 6.2, and 6.3. Far hem first, then near hem, then a loose end only if it shows loose hem, left gripper before right gripper on each edge.
- **Coaching note:** far, near, then ends only if they need it. Left bite, then right bite.

Violation: Loose hem left out

- **Visible cue:** Step 7 begins with hem hanging outside a side face or an end face, or with hem lying on the top face, and no bite is made there.
- **SOP rule broken:** Steps 6.1, 6.2, and 6.3. The hem is tucked under the bottom edge all the way round, with a second bite where the check finds hem still showing.

- **Coaching note:** walk your eye round all four sides before you call it done. Loose hem is the one thing the camera always sees.

Violation: Wrong phase order

- **Visible cue:** the tension pull is made before all four corners are hooked. A stroke is made before the tension pull. A bite is made before both halves are smoothed. Or a phase is skipped.
- **SOP rule broken:** Steps 2 to 6. Hook, then tension, then smooth, then tuck, each finished and checked before the next starts.
- **Coaching note:** hook, tension, smooth, tuck. Smoothing before the sheet is taut only moves the slack around.

Violation: Cloth let go in the air

- **Visible cue:** a gripper opens with cloth still lifted and not resting, so a corner, a hem, or part of the sheet drops. Or a gripper opens over the floor.
- **SOP rule broken:** Steps 1 to 6. A gripper opens only with the cloth resting on the mattress or with the hem already under the bottom edge.
- **Coaching note:** lower it until it rests, then open. Nothing is dropped.

Violation: Sheet lifted off or pulled off the mattress

- **Visible cue:** the whole sheet is lifted clear of the top face after Step 1. A corner or an edge is pulled so far that part of the sheet hangs off the mattress onto the floor, or the sheet slides so it hangs off one side. Or a gripper drags the sheet across the top face at any time other than a stroke.
- **SOP rule broken:** Steps 2 to 6. The sheet is worked on the mattress, a corner is lifted only as far as it takes to hook it, and nothing is dragged.
- **Coaching note:** the sheet stays on the bed. Lift a corner, not the sheet.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: a loose pocket, a popped corner, slack in the top, a wrinkle, hem still showing, or a shifted mattress.
- **SOP rule broken:** Steps 1 to 6.3. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** hooks a right corner, pulls the right end, strokes the right half, or bites on the right half. The **right gripper** hooks a left corner, pulls the left end, strokes the

left half, or bites on the left half. A gripper hooks a corner while the same gripper is meant to be pinning. Or cloth is handed from one gripper to the other.

- **SOP rule broken:** Steps 1 to 6. The **left gripper** works the left corners, the left end, and the left half. The **right gripper** works the right corners, the right end, and the right half. Nothing changes hands.
- **Coaching note:** left arm left, right arm right, and nothing changes hands.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a pocket unhooked, popped, or loose, the top wrinkled or slack, hem hanging outside a side face or lying on the top face, the mattress shifted, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 7. Confirm the corners, the top face, the hem, the mattress, the frame, and the base. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read a far corner or the far edge**, so whether a far pocket is hooked and seated or the far hem is tucked cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Mattress will not stay put:** a mattress that slides on the frame under a correct hook or a correct tension pull.
- **Mattress corner will not hold a pocket:** a corner so soft or so rounded that a correctly hooked pocket slides off on its own.
- **Bed blocks the reach:** a rail, headboard, footboard, or frame lip that stops a gripper reaching the top of a corner, a side face, or the bottom edge from the park.
- **Defective sheet:** elastic so slack that a seated pocket lets go on its own, a pocket mouth that will not stand open, an open seam, a tear, or a sheet that does not fit this mattress.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so a far corner, an end, the far hem, or the bottom edge cannot be reached without extending or folding the arm, or without reaching into the other arm's lane.

Annotation subtasks (from SOP)

1. Check and square the spread sheet
2. Pin the sheet near the middle of one half
3. Hook the far-left corner pocket
4. Hook the far-right corner pocket
5. Hook the near-left corner pocket
6. Hook the near-right corner pocket
7. Pull both ends outward, down, and under to tension the sheet
8. Stroke one half of the top face from the middle to its edges
9. Bite the far hem under on one half
10. Bite the near hem under on one half
11. Bite a loose end hem under
12. Return both arms home and end the episode